PUSL3190 Computing Project

Project Proposal

AI-Powered Leg Fracture Diagnosis System

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Table of Contents

Chapter 1: Problem Statement 2
Chapter 02: Project Description 3
Project Overview 3
Project Objectives 3
Specific objective: 3
Scope 5
Inclusions 5
Exclusions 5
Expected Outcomes 5
Keywords 5
Related Areas 6
Chapter 03: Research Gap 7
Identified Gap 7
Knowledge Gap 7
Technology Gap 7
Methodology Gap 7
Contextual Gap 7
How This Project Addresses the Gap 8
Chapter 04: Requirements Analysis 9
Functional Requirements 9
Non-Functional Requirements 9
User Requirements 9
Chapter 05: Finance/Budget Plan 10
Chapter 06: External Organizations/Support 11
Chapter 07: Time Frame / Timeline 12
Chapter 08: References 13

Chapter 1: Problem Statement

Bone fracture is one of the most common injuries treated in hospitals, with leg fractures comprising approximately 40% of the cases. Accuracy and timeliness in fracture diagnosis are important because prolonged exposure in identifying fractures, especially subtle or hairline ones, may result in long-term complications, inappropriate treatment, and increased healthcare costs. Access to competent radiologist is limited in many institutions, especially in developing countries such as Sri Lanka. Furthermore, the emergency department has a high volume of visits for assessment and has significant bottlenecks, with average wait times of 2–6 hours before a radiologist has been seen, decreasing patient outcomes, particularly in cases of trauma who may require immediate assistance.

Studies have shown that in emergency department fractures - as high as 30–50% - are either misdiagnosed or missed entirely in the initial assessments. Misdiagnosis may be exacerbated during night shifts, weekends, or if you happen to be in a rural hospital facility, where experience radiologist coverage is limited. Accurate detection of a fracture is essential for leg injuries as the femur, tibia, and fibula bones must bear the body weight and significantly contribute to mobility. Missed or misclassified leg fractures, will result in permanent deformities, development of infections, and extended recovery timetables.

Conventional diagnosis relies extensively on human reading of the X-ray, which is subjective and can potentially lead to errors with fatigue. While AI-based diagnostic tools are becoming available, either the prices are too high or the tools address only certain body parts, such as the wrist or right hip. There is a significant gap of low-cost diagnostics for leg fractures for example that is accessible in low-resource environments.

Therefore, we propose the development of an AI-Powered X-Ray Leg Fracture Detection System that utilizes deep learning using CNNs-based image processing for automated detection and diagnosis of bone fractures in the leg. With advances in machine vision and artificial intelligence, such systems show promise in improving diagnostic speed, accuracy and reliability depending less on a specialist and creating a solution that is less expensive, easier to use, and low-cost implementation in hospitals, diagnostic clinics, and emergency departments.

Chapter 02: Project Description

Project Overview

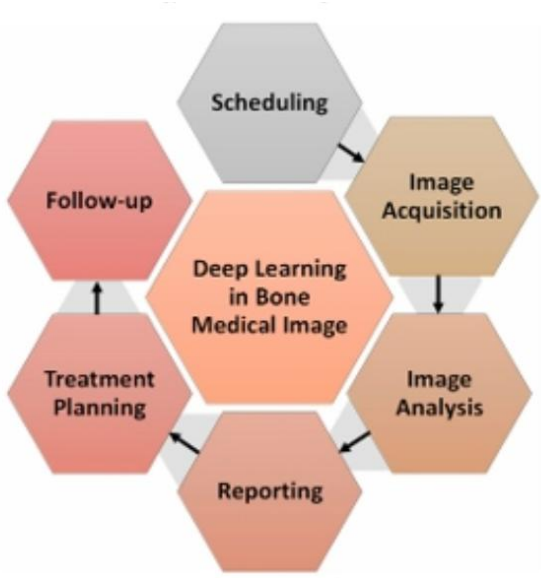
This initiative aims to build an intelligent, cloud-based diagnostic system that utilizes dto deep learning using CNNs detect leg fractures from X-ray images. The system will assess uploaded X-ray images, assess the presence and type of fracture, and use Grad-CAM heatmaps as a tool for how it visually represented.

The goal is to create a diagnostic tool that can provide medical professionals with near real-time diagnostic feedback, while limiting the need for specialist radiologists in rural or under-staffed hospitals. This would significantly reduce the time delays and human errors in fracture diagnosis, especially in settings where radiologists are scarce and overburdened. By automating the initial detection process, the system can assist healthcare workers in rapidly triaging patients, prioritizing those with severe fractures for immediate care.

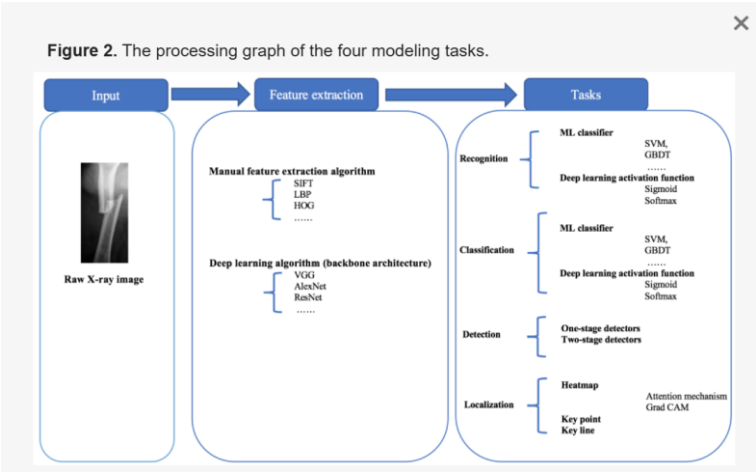
Project Objectives

Specific objective:

1. Design and train a CNN-based deep learning model using transfer learning to classify X-ray images into categories:
    - Normal
    - Femur Fracture
    - Tibia Fracture
    - Fibula Fracture
    - Multiple Fractures
2. Achieve a minimum accuracy of **90%** and recall of **95%** to minimize missed fractures.
3. Implement **Grad-CAM visualization** for interpretable fracture localization.
4.Provide automated severity estimation (mild / moderate / severe).
5. Build a user-friendly web application with drag-and-drop X-ray upload and instant diagnostic output.
6. Store diagnostic history securely for analysis and reporting.



"Figure 1:1: Workflow of deep learning in bone medical imaging, from scheduling and image acquisition to reporting and follow-up."



“Figure 1:2: The processing graph of the four modeling tasks”

Scope

The system focuses exclusively on the detection of leg fractures from X-ray images. It will not include MRI or CT scan analysis, nor will it provide full medical treatment recommendations. The project will prioritize accuracy, interpretability, and ease of use for medical staff.

Inclusions

- Module for Uploading X-ray Images: and enables users to upload leg X-ray images in all common image formats (JPEG, PNG).
- AI-Based Detection Engine: automatically detects leg fractures using deep learning (CNNs).
- Classification System: detects and classifies fractures as Normal, Femur Fracture, Tibia Fracture, Fibula Fracture, or Multiple Fractures.
- Grad-CAM Visualization: generates color-coded heatmaps light up fracture components.
- Authentication and Security: includes a secure log-in system with JWT authentication, and encrypted data storage.
- Web-Based Interface is accessible via web browser without installation, offering a drag-and-drop upload option for images

Exclusions

- MRI or CT Scan Analysis – limited to X-ray imaging only
- Full Medical Treatment Recommendation – the system provides diagnostic support, not prescriptions.
- 3D Image Reconstruction – focuses solely on 2D X-ray processing.
- Mobile Application Version – current implementation is web-based only.
- Offline Functionality – requires an internet connection for cloud-based model inference.

Expected Outcomes

1. **Rapid diagnosis:** Reduce X-ray interpretation time from hours to seconds.
2. **Improved accuracy:** Achieve ≥90% diagnostic accuracy for leg fractures.
3. **Accessibility:** Provide affordable diagnostic support to rural hospitals.
4. **Transparency:** Enable explainable AI through heatmap visualization.

Keywords

- Artificial Intelligence in Healthcare
- Convolutional Neural Networks
- Medical Image Processing
- X-ray Image Classification
- Explainable AI (Grad-CAM)
- Diagnostic Assistance System

Related Areas

- Medical Imaging and Radiology AI
- for Biomedical Applications
- Computer-Aided Diagnosis (CAD) Systems
- Machine Vision in Healthcare
- Image Recognition and Classification
- Cloud-Based Healthcare Solutions
- Predictive Analytics in Medicine
- Human–Computer Interaction (Healthcare Interfaces)

Chapter 03: Research Gap

Numerous studies have utilized deep learning and convolutional neural networks (CNNs) for different X-ray-based diagnostic purposes. For instance, a 2020 study, “Femoral neck fracture detection in X-ray images using deep learning and genetic algorithm approaches” (Beyaz, 2020), detects fractures of the femoral neck region utilizing frontal pelvic X-rays.

In chest imaging, automated detection of diseases on chest X-rays is now commonplace. For example, the study “Deep Learning-Based Object Detection Strategies for Disease Detection and Localization in Chest X-Ray Images” (Cheng) illustrates how deep learning is used for disease detection in the general thoracic population.

A study, “Use of deep learning methods for hand fracture detection from plain hand radiographs” (publication, 2024) , applied CNNs and transfer learning to obtain a high accuracy (≈93%) of distinguishing fractured and normal hand and wrist radiograph

A comprehensive systematic review (Skeletal Fracture Detection with Deep Learning: A Comprehensive Review, 2023) summarized 40 recent studies on fracture detection of X-rays, and reported that most studies examined wrist, elbow or general skeletal sites rather than leg fractures.

Identified Gap

Knowledge Gap

Most of the current research is centered on fracture identification in the hand, wrist, or chest. There is limited research on detection of leg fractures (femur, tibia, fibula). There appears to be a lack of research specifically focused on detection of lower limb fractures using X-ray images.

Technology Gap

Existing deep learning models or tools are not catered towards leg fractures, and likely require complex infrastructure or a specialist. There are no lightweight web-based options available for use in lower resourced medical institutions.

Methodology Gap

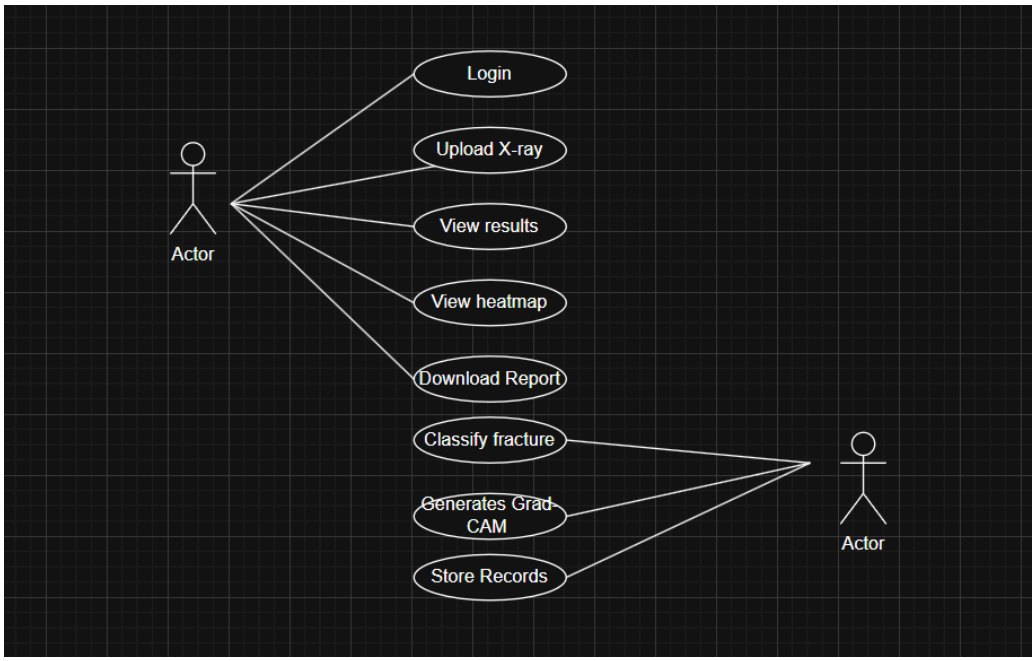
Most methods simply provide a classification outcome without any visual interpretability. Very few studies have been done that utilize Grad-CAM or explainable AI techniques to localize fracture area(s) or visualize diagnoses in real time for the clinician.

Contextual Gap

Most studies engaged previously have done so at a high resource level of medical care, which is not applicable to low-resource settings, or even applicable to a country like Sri Lanka, where medical resources and access to radiologists or diagnostic settings are limited.

How This Project Addresses the Gap

- Explicitly focusing on long bone fractures of the leg (femoral, tibia, fibula) when much of the previous research was on other parts of the body (hands, wrist, chest).
- Assembling or building a fracture dataset specifically for the leg area, training a CNN-based deep learning model designed for the long-bones of the leg.
- Developing a visualisation using Grad-CAM heatmaps to enhance interpretability where the clinician can improve their trust of the model through a visual representation of the fracture location.
- Creating a web-based image diagnostic tool being offered with drag-and-drop upload and web-based interface for usability and expedite feedback system - ideal for hospitals and medical clinics with limited fission of specialists.



(Figure1:3)

The use case diagram shows the interaction between the User and the Internal/Automatic System.
Users can log in, upload X-ray images, view diagnosis results and Grad-CAM heatmaps, download diagnostic reports, view their diagnosis history, and log out.
The internal system automatically classifies leg fractures, generates Grad-CAM heatmaps, and stores diagnostic records to support accurate and efficient diagnostics.

Chapter 04: Requirements Analysis

Functional Requirements

- User Authentication: Role-based secure authentication (medical personnel, administrator).
- X-ray Upload: Upload leg X-ray images (JPEG/PNG).
- Fracture Classification: Classification using CNN (Normal, Femur, Tibia, Fibula, Multiple).
- Severity Classification: Mild / Moderate / Severe.
- Grad-CAM Visualization: Visualize fracture locations.
- Downloadable Reporting: Download diagnostic reports.
- History Management: Store and find past X-rays and predictions.

Non-Functional Requirements

- **P**erformance: Takes ≤ 10 seconds for diagnosis; allows ≥ 20 concurrent users.
- Accuracy: Classification accuracy ≥ 90%; recall ≥ 95%.
- Usability: Web interface that is intuitive; allow user to drag-and-drop files for upload.
- Security: JWT (JSON Web Token) authentication; encrypted storage; HTTPS (Hyper Text Transfer Protocol Secure) communication.
- Reliability: Can handle low-quality or corrupted images; automatically restarts after failure.
- Scalability: Based on cloud technologies, easy to expand for new fracture types.
- Maintainability: Built into modular architecture; documentation available.

User Requirements

- Quick access and upload: Users can easily log in and upload X-ray images in common formats.
- Instant feedback and visualization: Users receive immediate fracture diagnosis with Grad-CAM heatmap.
- Secure storage and history: Users' images and diagnostic results are stored safely and can be retrieved anytime.

Chapter 05: Finance/Budget Plan

Item	Estimated Cost (LKR)	Description
Domain/Hosting	22,000	Web app deployment, scalable cloud storage with secure access
GPU/Hardware	80,000	Cloud-based GPU services for model training and inference
Cloud Services	30,000	Model inference, backups, and data storage services with pay-as-you-go
Dataset Acquisition	0	Use of publicly available datasets from Kaggle and other open sources
Software Tools	15,000	Open-source libraries and frameworks
Maintenance & Misc	8,000	Ongoing updates, patches, monitoring, and energy costs

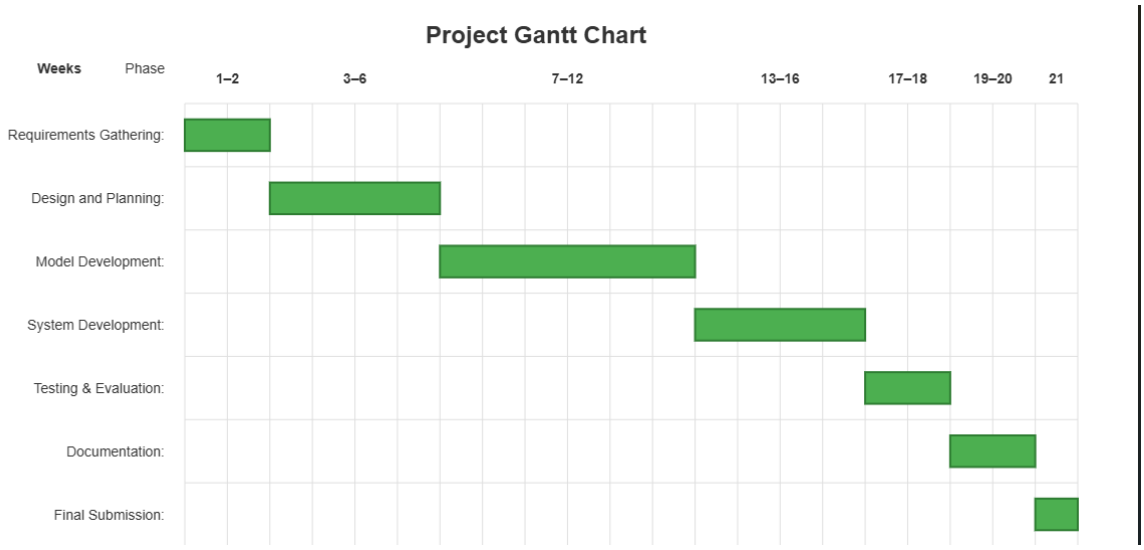
The overall estimated budget accommodates project basic necessities, including hosting and GPU cost, cloud services, software tools, and maintenance. Cost will be minimized through the use of publicly available datasets while maintaining a proper process for training, deploying, and long-term viability.

Chapter 06: External Organizations/Support

1. Kaggle Bone Fracture Dataset
   A large collection of annotated X-ray images focusing on bone fractures, including tibia and fibula fractures, suitable for training and validation of AI models.
2. FracAtlas Dataset
   A musculoskeletal radiograph dataset with over 4,000 X-ray images with detailed annotations for fracture classification, localization, and segmentation, curated for deep learning tasks. Available under a CC-BY 4.0 license.
3. Mendeley X-ray Bone Fracture Dataset
   A clinically curated dataset focusing on simple and comminuted fractures with high-quality, augmented X-ray images to enhance model robustness.
4. GRAZPEDWRI-DX Dataset (available via GitHub)
   Contains over 20,000 annotated pediatric wrist radiograph images, useful for fracture detection algorithm development, especially in extremity fractures.
5. Roboflow X-ray Bone Fracture Dataset
   An open dataset with annotated images and a pre-trained fracture detection model API, providing a resource for rapid prototyping and testing.

Chapter 07: Time Frame / Timeline

Phase	Weeks	Tasks
Requirements Gathering	01–02	Literature review, stakeholder consultations, finalize project requirements
Design and Planning	03–06	Design architecture, wireframes, model selection, dataset sourcing
Model Development	07–12	Data preprocessing, CNN training, Grad-CAM development
System Development	13–16	Backend/front-end implementation, API and UI development, security integration
Testing & Evaluation	17–18	Usability testing, model validation, user feedback, refinements
Documentation	19–20	Final report, user guide, presentation materials
Final Submission	21	Submit deliverables and present final demo



(figure1:4)

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